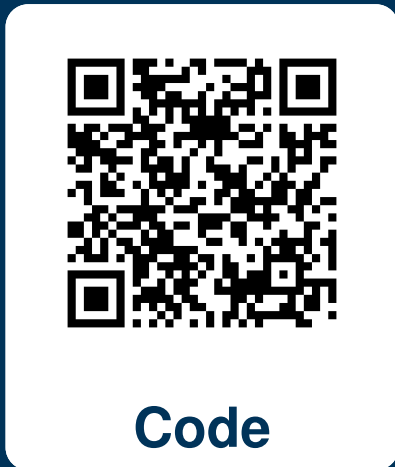


Can a Vision-Language Model Group 2D Masks into 3D Objects?

VLM-based edge scoring and geometric-semantic fusion for 3D instance segmentation

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Code

TUM

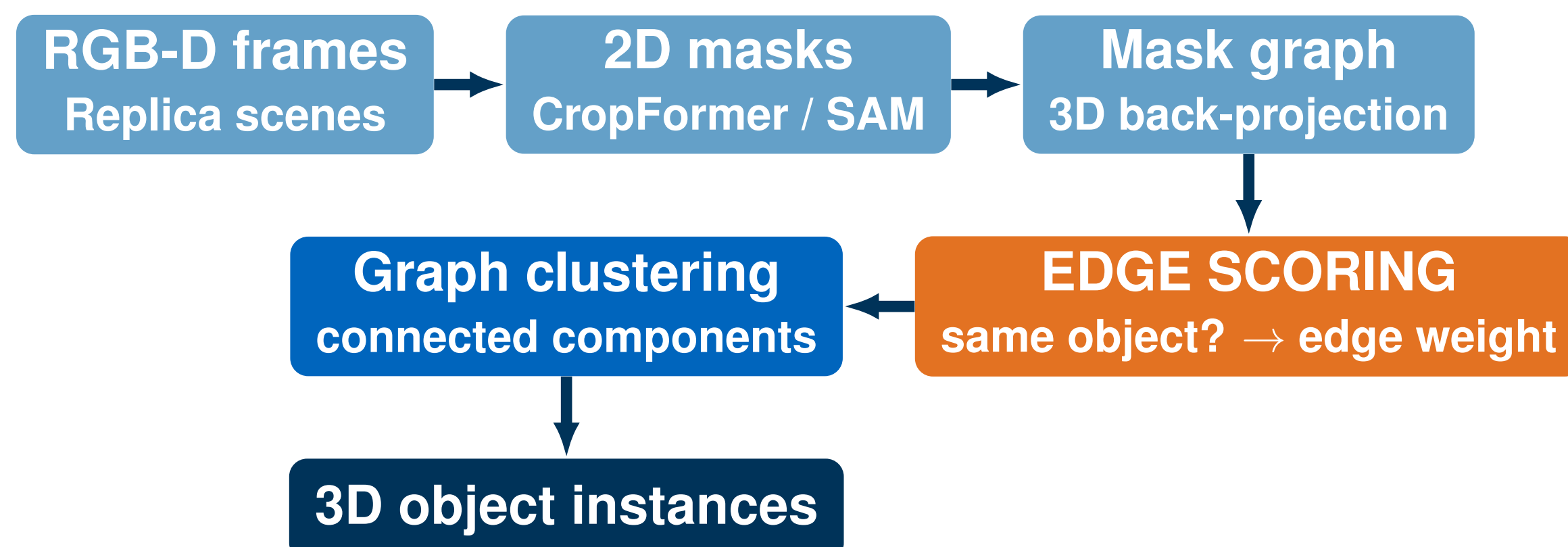
Motivation



Class-agnostic 2D masks (CropFormer) in one RGB-D frame of Replica room0.

- Goal: group per-frame **2D instance masks** into consistent **3D objects**.
- MaskClustering (CVPR'24) solves this **purely geometrically** — no semantics.
- Can a vision-language model (**Qwen3-VL**) judge “same object?” instead — or in addition?
- **We ablate two axes:** the 2D mask backbone (CropFormer vs SAM 1/SAM 3) and the edge scorer (geometry → VLM → **learned fusion**).

Method: One Pipeline, Three Edge Scorers



Masks become **graph nodes**; candidate pairs ($3D\ overlap \geq \tau$) get an edge if the **edge score** passes a threshold. Connected components form 3D objects. **We keep the pipeline fixed and swap only the edge scorer.**

Edge Scoring: Three Approaches

1 · GEOMETRIC (baseline)

View consensus: a mask pair is merged if enough frames that observe both also contain them inside one mask (*supporters* / *observers*). Purely geometric — fails on adjacent objects with high 3D overlap.

2 · SEMANTIC (Qwen3-VL)



What the VLM sees: two views, one region outlined each (A red, B blue).

The VLM answers “**same physical object?**” with a confidence score, used directly as the edge weight. **Zero-shot** it is over-conservative (too few edges, fragmented scenes); **LoRA finetuning** on 5 held-out Replica scenes sharpens it, but raw VLM edges stay too sparse to group instances alone.

3 · LEARNED FUSION (MLP)

A small MLP (2×32 , $\sim 1.3k$ params) combines **geometric features** (view consensus, #observers, 3D IoU) with the **Qwen confidence** into one edge probability. Trained on 6 scenes, validated on 1, tested on *room0*.

Question: does semantics add anything on top of geometry? **On Replica, barely — geometry dominates.**

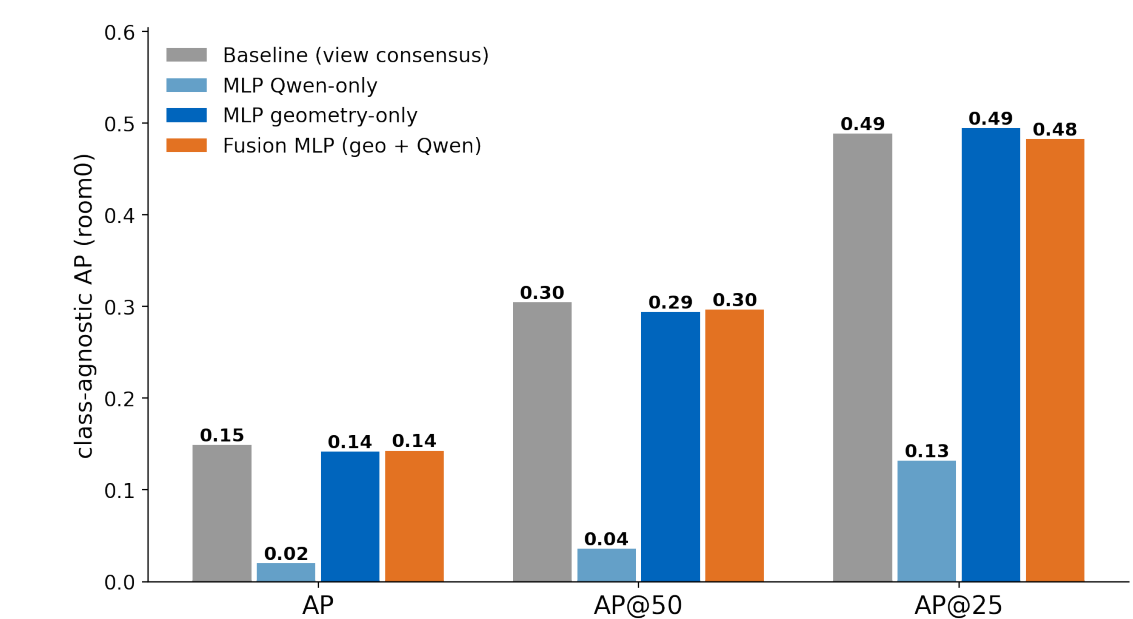
Summary & Outlook

- **CropFormer is the strongest mask backbone (M2):** it beats SAM 1 and SAM 3 on downstream 3D AP, so all later milestones build on CropFormer masks.
- **The raw VLM signal is weak:** zero-shot and finetuned Qwen edges give $AP \approx 0.02$ — too sparse to group 3D instances on their own.
- **Geometry drives the fusion MLP:** it reaches baseline-level AP, but adding the Qwen score moves AP by < 0.01 — semantics adds little on top of geometry on Replica.

KEY FINDING The fusion MLP matches the geometric baseline but does not surpass it: on Replica the VLM contributes almost no signal once view-consensus geometry is in the model.

NEXT STEPS Stronger VLM supervision · projection-based single-image prompting · more scenes and larger VLMs.

Results — Replica room2 (class-agnostic AP)



Edge scorer	AP	AP@50	AP@25
View consensus (baseline)	0.082	0.189	0.423
Qwen finetuned (LoRA)	0.018	0.033	0.173
Average (Qwen + consensus)	0.075	0.191	0.430
MLP Qwen-only	0.018	0.033	0.173
Fusion MLP (geo + Qwen)	0.083	0.205	0.441

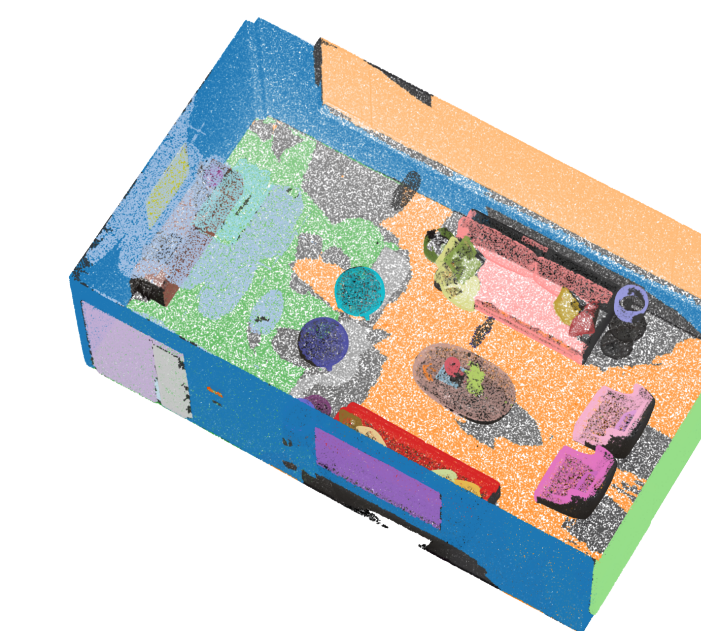
Leak-free test scene (Qwen v2 fine-tuned on office0/office1/room0 only). The fusion MLP marginally edges the baseline ($AP@50\ 0.205$ vs 0.189) but does not clearly beat it; Qwen-only and naive averaging stay at or below geometry. All variants share the same clustering and CropFormer masks.

Mask backbone (M2)	AP	AP@50	AP@25
CropFormer	0.149	0.305	0.489
SAM 3 (conf 0.25)	0.123	0.224	0.370
SAM 1 (ViT-H)	0.061	0.107	0.222

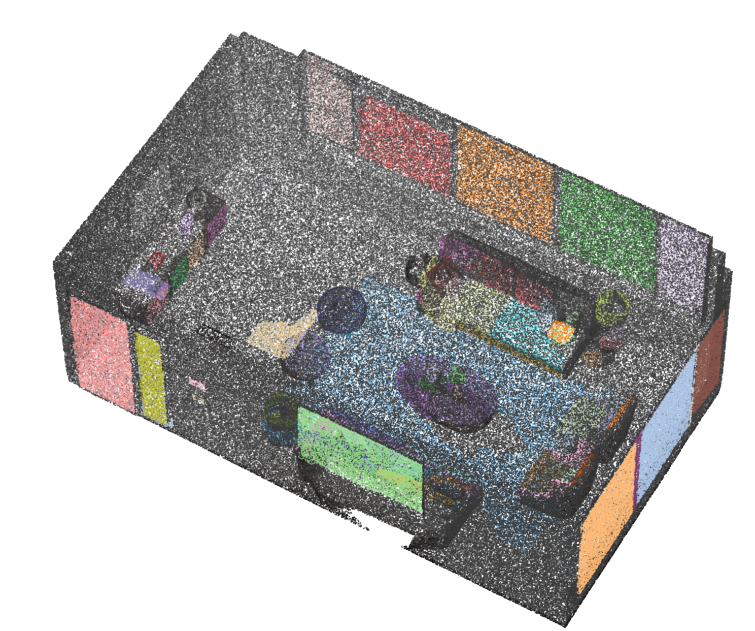
Same geometric pipeline, different 2D masks. CropFormer's class-agnostic masks give the best 3D AP, so all later milestones use them.

Predicted 3D Instances — room2

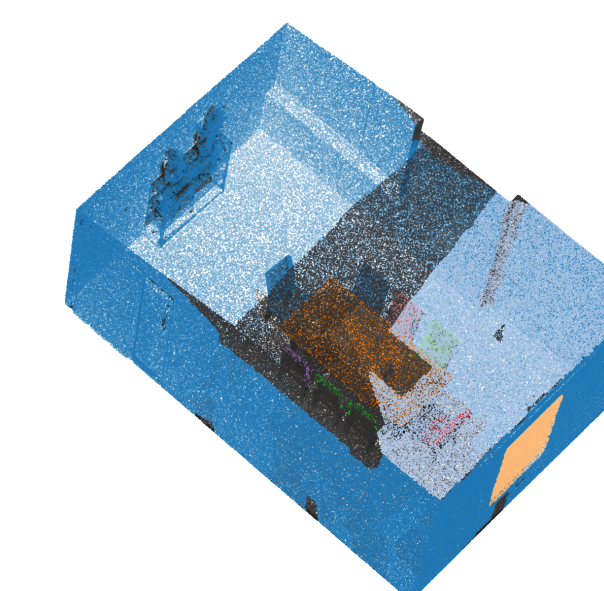
Baseline (CropFormer)



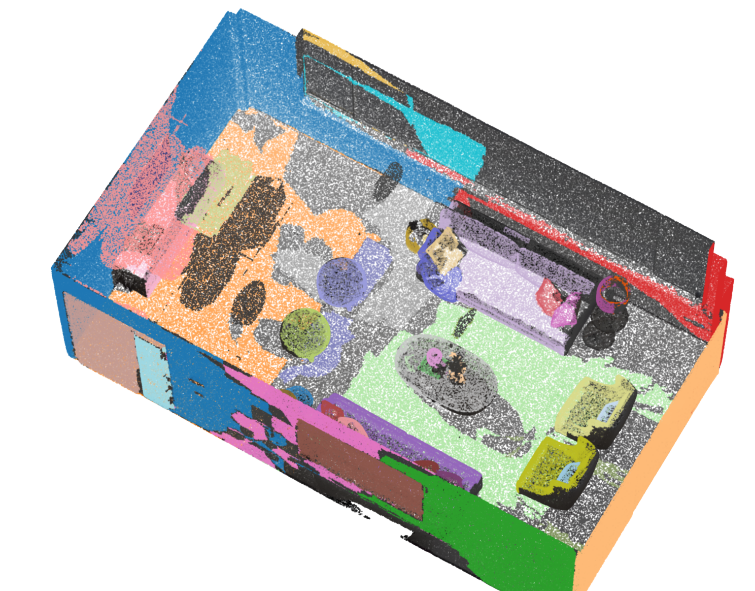
SAM 3 masks



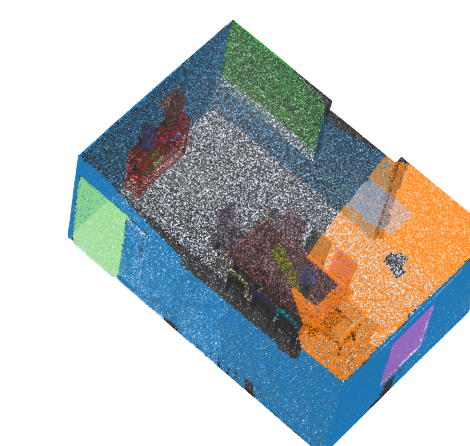
Qwen finetuned



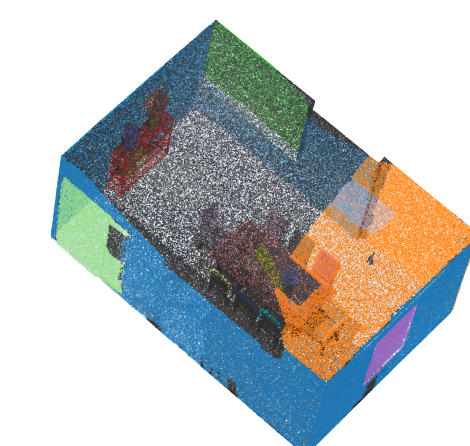
Fusion MLP



Baseline (zoom)



Fusion MLP (zoom)



One color per predicted instance, same room2 viewpoint. **Top:** full scene, four methods — SAM 3 over-segments surfaces into slabs, CropFormer + geometry recover cleaner objects. **Bottom:** close-up where baseline and fusion differ.